Authors: Annika Bürger*, Martin Dugas

Institute of Medical Informatics, University of Münster, Germany

Institut für Medizinische Informatik

IMI

Visualization supported association analysis of genomic ranges annotations

*corresponding author e-mail annika.buerger@ukmuenster.de

Introduction

With NGS there arise plenty genomic data. Lists of genomic ranges annotated with scores, names and other values are often explored only partially to answer specific scientific questions. Unknown dependencies between annotations can therefore stay undiscovered. In a NGS-data set, for example from the 1000 genomes project, there are not only millions of reads, but also normally >10 phenotypic attributes. What implies more than $\binom{10}{2}$ possible comparisons of pairs of attributes which suggests an systematic and automatic approach to analyse the data.

Methods

We developed a method to examine associations between all possible pairs of annotations. For this we first categorize and summarize the annotations according to their scale.

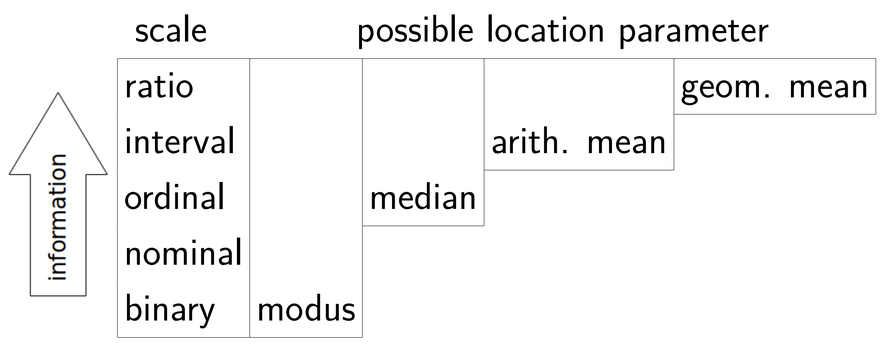

Figure 1: Scales of data in order of information content with possible location parameter

Differently scaled data provides a different amount of information and can be converted from scales which much information to scales with less information. The scale *binary* can be treated as one instance of scale *nominal*. Scales *ratio* and *interval* as metric scales contain numbers, the other scales are categorical. To get an impression of the distribution of an annotation, the location and the dispersion ot the data play an important role.

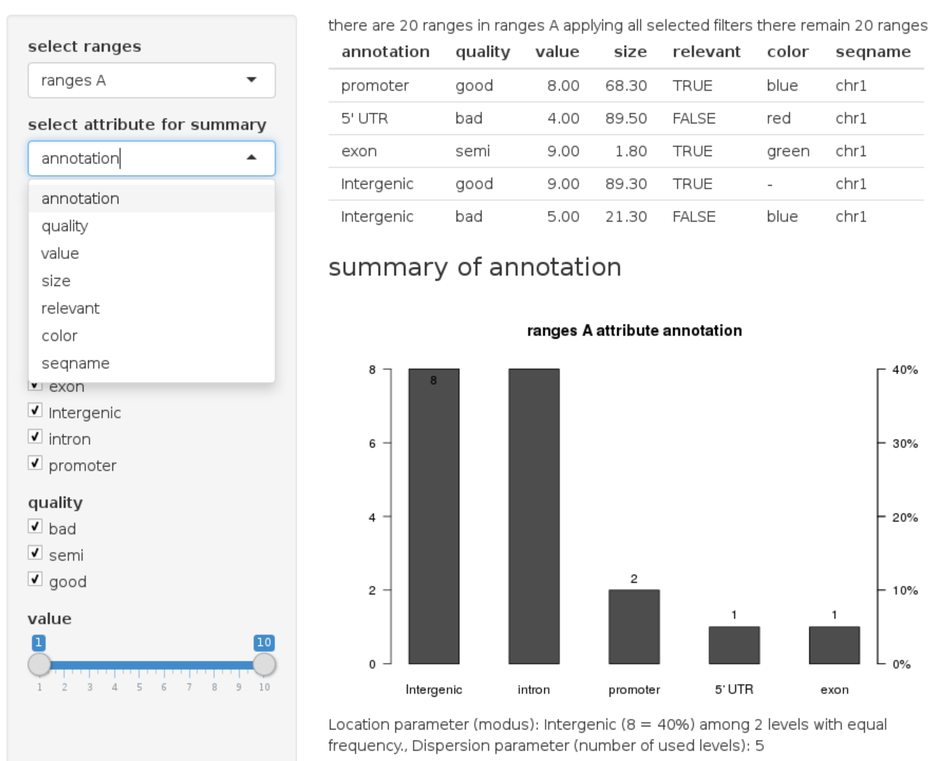

Figure 2: Selecting ranges and attributes for summary

For implementation we choose R. In R/Bioconductor annotated genomic ranges can be easily handled as *GRanges* objects [1] and implementations of statistic models can be used to determine associations between annotations. The user interface is implemented with the R package *Shiny* [2]. To visualize categorical data ordered or unordered barplots are available. For metric data histograms, boxplots or for example violin-plots are suitable. In a second step an adequate statistic model is chosen to find possible dependencies between the annotations. The chosen model depends on the scale of the compared annotations. There are a lot of models and tests available, for example Pearson correlation, t-test, Wilcoxon-rank sum test, Kruskal-Wallis test, fisher exact test and many more.

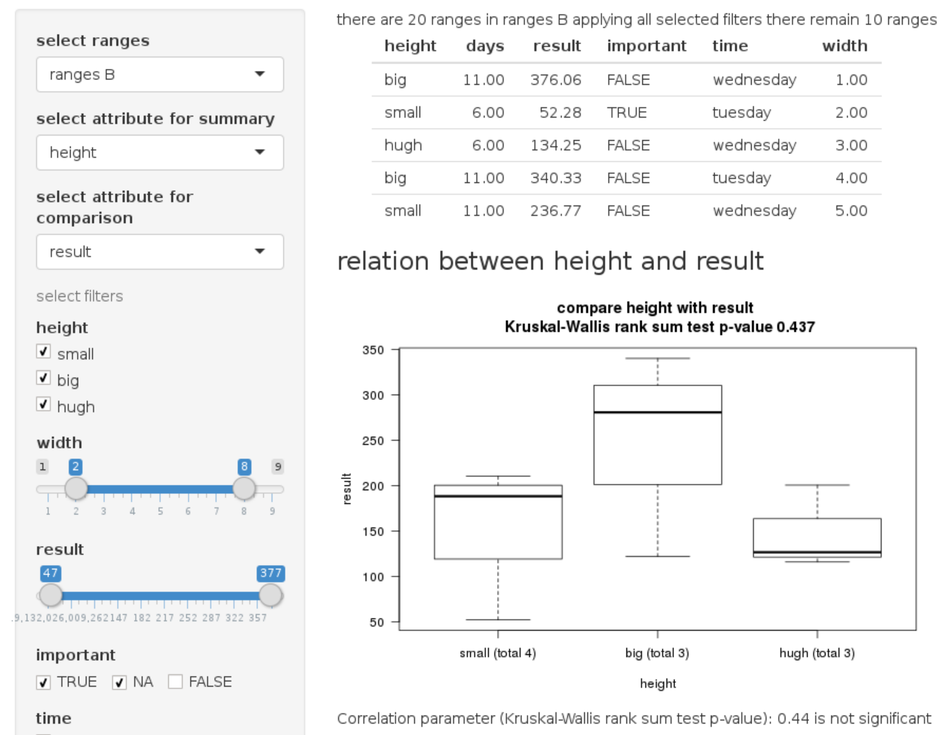

Figure 3: Selecting filters and annotations for comparison

To facilitate the differentiation between significant dependencies and random effects the results are then visualized again depending on scales, for example scatterplots and grouped boxplots. Furthermore it is possible to filter data by all possible annotations.

Results

By the categorization of annotation data in different scales it is possible to automatically compare all pairs of annotation data with each other. With adequate visualisation and descriptive tests on the data it is further possible to rank the results for further investigation.

Outlook

As a further feature it should be possible to automatically compare two or more sets of annotated genomic ranges with each other. For this it is necessary to link ranges in one set with ranges in another set, but this requires deeper investigation.

References

[1] Lawrence M, Huber W, Pagès H, Aboyoun P, Carlson M, et al. (2013) *Software for Computing and Annotating Genomic Ranges*. PLoS Comput Biol 9(8): e1003118. doi:10.1371/journal.pcbi.1003118 **[2]** Winston Chang, Joe Cheng, JJ Allaire, Yihui Xie and Jonathan McPherson (2017). *shiny: Web Application Framework for R. R package version 1.0.3.* https://CRAN.R-project.org/package=shiny

UKM Universitätsklinikum Münster

WWU MÜNSTER

medizinische fakultät Westfälische Wilhelms-Universität Münster